

CH102 Practice Exam #1

WAY longer than actual exam, answers are at the end

True/False

Indicate whether the sentence or statement is true or false.

- ____ 1. Organic chemistry can be defined as the chemistry of carbon-containing compounds.
- ____ 2. Organic compounds can contain other elements such as oxygen or nitrogen.
- ____ 3. Organic compounds are always derived from a living source.
- ____ 4. An unlimited number of carbon atoms can bond together to form chains.
- ____ 5. Most organic compounds contain ionic bonds.
- ____ 6. There are more inorganic compounds than organic compounds.
- ____ 7. Most inorganic compounds have a higher melting point than organic compounds.
- ____ 8. Inorganic compounds are never found in living systems.
- ____ 9. Isomers differ from each other in the number of neutrons present in each.
- ____ 10. Isomers always have the same empirical (molecular) formula.
- ____ 11. Isomers have identical physical and chemical properties.
- ____ 12. Carbon-carbon bonds are usually strong and covalent.
- ____ 13. A characteristic of the alkanes is that they burn in air.
- ____ 14. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2$ is the formula for a saturated hydrocarbon.
- ____ 15. Alkenes must have at least two carbon atoms.
- ____ 16. Alkenes only have one double bond.

____ 17. The addition of bromine (Br_2) to an alkene results in an alkane because one bond of the multiple bond is broken.

____ 18. One of the halogenation reactions occurs when a halogen, a member of group VIIA, reacts with alkene.

____ 19. Cyclic compounds do not undergo halogenation reactions.

____ 20. Markovnikov's rule states that when a hydrogen and a different molecule (OH^- , Br^-) add to a double bond the hydrogen goes to the end of the double bond that already had more hydrogens

____ 21. An alkene with one multiple bond can be converted to an alkane by hydration.

____ 22. Polymers are compounds that are composed of repeating units chemically bound to each other.

____ 23. The chemical properties of alkynes are very different from those of alkenes.

____ 24. 2-butyne can exist as *cis*- and *trans*- isomers.

____ 25. Two moles of hydrogen gas would be required to convert one mole of 2-butyne into butane.

____ 26. Benzene is an alkene with more than one multiple bond.

____ 27. Phenyl is the name given to benzene when it is a substituent of a larger molecule.

____ 28. Aromatic compounds dissolve well in a nonpolar solvent.

Multiple Choice

Identify the letter of the choice that best completes the statement or answers the question.

- ____ 29. What is the reason why there are so many organic compounds?
- Organic compounds are produced by living things.
 - There are cyclic organic compounds, and none of the inorganic compounds are cyclic.
 - Carbon atoms can form stable covalent bonds with other carbon atoms.
 - Organic compounds can be larger than inorganic compounds.

____ 30. Which applies to the manner in which carbon bonds in alkanes?

- a. The bonds are formed with carbon using an sp^3 configuration.
- b. The bonds between carbon and other atoms are ionic.
- c. carbon to carbon bonds are expected to be double bonds.
- d. The bonds formed by carbon all use p orbitals.

____ 31. Which statement referring to functional groups is correct? Functional groups are

- a. where nothing happens in a chemical reaction
- b. the parts of molecules used to classify organic molecules
- c. where most chemical reaction occur
- d. both b and c

____ 32. Identify the correct statement: The alkanes represent one family of organic compounds composed of carbon compounds

- a. in which the halogens are a major component
- b. in which there is at least one double bond between carbons
- c. that are composed only of carbon and hydrogen
- d. in which hydrogens can be bonded to other hydrogens

____ 33. Which of the following formulas represents an organic compound?

- a. CH_4
- b. $CH_4CH_2CH_3$
- c. $CH_3CH_3CH_3$
- d. they all do

____ 34. Which of the following is an alcohol?

- a. CH_3CH_2COOH
- b. CH_3CH_2OH
- c. CH_3COCH_3
- d. CH_3CH_2Br

____ 35. What is meant by the term "saturated" when describing hydrocarbons?

- a. The hydrocarbons are saturated with water and are wet.
- b. The hydrocarbons are saturated with carbons in its parent chain.
- c. The hydrocarbons are saturated with hydrogens and have only single bonds between carbons.
- d. The compound is pure

____ 36. The naming of organic compounds is specific. Which is an incorrect statement?

- a. The smaller of the compounds have been known long enough to have common names.
- b. The naming system of organic compounds is set out

by IUPAC rules.

- c. The naming system for hydrocarbons requires the identification of the longest carbon chain.
- d. The naming system for organic compounds requires and IUPAC name and a common name.

____ 37. Identify the correct statement. Cycloalkanes

- a. can have as few as two carbons.
- b. are unsaturated hydrocarbons.
- c. have more hydrogens than straight chain alkanes with the same number of carbons
- d. these are all wrong

____ 38. Which of the following terms applies to the alkanes?

- a. hydrophobic
- b. hydrophilic
- c. soluble
- d. bases

____ 39. What is a characteristic shared by the alkanes?

- a. solids
- b. polar
- c. flammable
- d. colored

____ 40. Which of the following is a used for alkanes?

- a. fuel
- b. lubricant
- c. solvent
- d. all of the above

____ 41. Which of the following formulas represents a saturated hydrocarbon?

- a. C_2H_6
- b. C_2H_4
- c. C_3H_6
- d. C_5H_{10}

____ 42. All of the members of the alkane series of hydrocarbons have the general formula

- a. C_nH_{2n}
- b. C_nH_n
- c. C_nH_{2n+2}
- d. C_nH_{2n-2}

____ 43. Which of the following structural formulas represents an unbranched alkane?

- a. $\begin{array}{c} \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
- b. $\begin{array}{c} \text{CH}_3-\text{CH}_2-\text{CH}_2 \\ | \\ \text{H}_2\text{C}-\text{CH}_3 \end{array}$
- c. $\begin{array}{c} \text{H}_2\text{C}-\text{CH}_2 \\ | \\ \text{H}_2\text{C}-\text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$
- d. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$

____ 44. Which of the following compounds exhibits structural isomerism?

- a. propane
b. methane
c. butane
d. ethane

____ 45. How many structural isomers are possible for butane (molecular formula C_4H_{10})?

- a. 2
b. 3
c. 4
d. 5

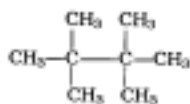
____ 46. Which of the following compounds is a structural isomer of 2-methylbutane?

- a. propane
b. w-methylpropane
c. butane
d. pentane

____ 47. A structural isomer of $\begin{array}{c} \text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$ is

- a. $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
- b. $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3$
- c. $\text{CH}_3-(\text{CH}_2)_5-\text{CH}_3$
- d. $\begin{array}{c} \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_2-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$

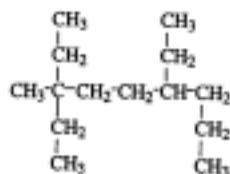
____ 48. The name for the following structure will end with



- a. ethane

- b. butane
c. hexane
d. octane

____ 49. Which is the longest carbon chain?



- a. 9
b. 6
c. 7
d. 10

____ 50. The IUPAC name for $\begin{array}{c} \text{CH}_3-\text{CH}-\text{CH}_2-\text{CH}_3 \\ | \\ \text{H}_2\text{C}-\text{CH}_3 \end{array}$ is

- a. 2-ethylbutane
b. 2-methylbutane
c. 3-methylpentane
d. 3-ethylbutane

____ 51. The compound $\begin{array}{c} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_2\text{CH}_2\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3 \\ | \\ \text{CH}_2\text{CHCH}_3 \end{array}$ has the IUPAC name

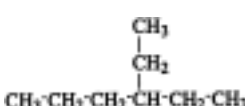
- a. 4-isopropyloctane
b. 5-isopropyloctane
c. 5-sec-butyloctane
d. 4-propyloctane

____ 52. The structural formula of 1-bromo-2,2-dimethylpropane is

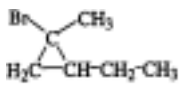
- a. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{CH}_2-\text{Br} \\ | \\ \text{CH}_3 \end{array}$
- b. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}-\text{CH}_2-\text{Br} \\ | \\ \text{CH}_3 \end{array}$
- c. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{Br}-\text{CH}_2-\text{CH} \\ | \\ \text{CH}_3 \end{array}$
- d. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}_2-\text{C}-\text{Br} \\ | \\ \text{CH}_3 \end{array}$

____ 53. A general formula for cycloalkanes is

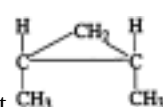
- a. C_2H_{2n+2}
- b. C_nH_{2n}
- c. C_nH_{2n-2}
- d. C_nH_{2n-4}

____ 54. The IUPAC name of  contains an ethyl group located at position

- a. 1
- b. 2
- c. 3
- d. 4

____ 55. A correct IUPAC name for  is

- a. 1-bromo-3-ethyl-1-methylcyclopropane
- b. 1-bromo-2-ethyl-1-methylcyclopropane
- c. 1-bromo-1-ethyl-2-methylcyclopropane
- d. 2-bromo-1-ethyl-2-methylcyclopropane

____ 56. The component  is ____-1,2-

- a. cis
- b. trans
- c. tis
- d. crans

____ 57. An alkane with 10 carbon atoms is a ____ at room temperature.

- a. solid
- b. liquid
- c. gas
- d. not possible to tell

____ 58. Alkanes are ____ in water and ____ than water.

- a. insoluble, less dense
- b. soluble, less dense
- c. insoluble, more dense
- d. soluble, more dense

____ 59. Complete combustion of pentane produces

- a. carbon monoxide and water
- b. carbon dioxide and water

- c. carbon dioxide and carbon monoxide
- d. carbon monoxide and carbon

____ 60. Alkanes react readily with

- a. sodium hydroxide
- b. hydrochloric acid
- c. sulfuric acid
- d. none of the above

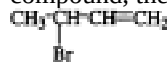
____ 61. Name a difference between a saturated and an unsaturated hydrocarbon.

- a. Saturated hydrocarbons are composed of only carbon and hydrogen, and unsaturated hydrocarbons include other atoms than just carbon and hydrogen.
- b. Saturated hydrocarbons do not contain multiple bonds between carbons, but unsaturated hydrocarbons do contain multiple bonds.
- c. Unsaturated hydrocarbons are flammable, but saturated hydrocarbons aren't flammable.
- d. Saturated hydrocarbons are essentially insoluble. Unsaturated hydrocarbons are soluble.

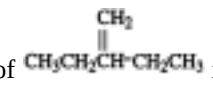
____ 62. What number would be used to indicate the double bond position in the IUPAC name for $CH_3CH_2CH=CHCH_3$?

- a. 1
- b. 2
- c. 3
- d. 4

____ 63. In the IUPAC name for the following compound, the -Br group is located at position



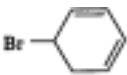
- a. 1
- b. 2
- c. 3
- d. 4

____ 64. The IUPAC name of  is

- a. 3-ethyl-1-butene
- b. 2-ethyl-2-butene
- c. 3-ethyl-3-butene
- d. 2-ethyl-1-butene

_____ 65. The IUPAC name of $\text{CH}_3\text{C}(\text{CH}_3)=\text{CH}-\text{CH}=\text{CH}_2$ is

- 2-methyl-1,4-pentadiene
- 2-methyl-2,4-dipentene
- 4-methyl-1,3-pentadiene
- 4-methyl-2,4-pentadiene

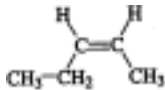
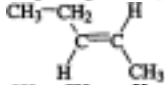
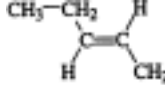
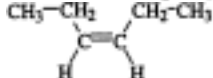
_____ 66. A correct IUPAC name for  is

- 5-bromo-1,3-cyclohexadiene
- 2-bromo-1,3-cyclohexadiene
- 2-bromo-1,4-cyclohexadiene
- 3-bromo-1,5-cyclohexadiene

_____ 67. Which of the following compounds could exist in the form of two geometric isomers?

- $\text{CH}_3\text{C}(\text{CH}_3)=\text{CH}-\text{CH}_3$
- $\text{CH}_3\text{C}(\text{Cl})=\text{C}(\text{Cl})\text{CH}_3$
- $\text{CH}_3\text{C}(\text{Cl})=\text{C}(\text{Cl})\text{CH}_3$
- $\text{CH}_3\text{C}(\text{Cl})=\text{C}(\text{Cl})\text{CH}_3$

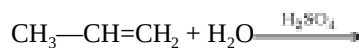
_____ 68. Which of the following compounds is *trans*-3-hexene?

- 
- 
- 
- 

_____ 69. Which of the following represents an addition reaction?

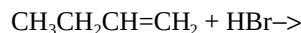
- $\text{HX} + \text{C}_4\text{H}_8 \rightarrow \text{C}_4\text{H}_9\text{X}$
- $\text{X}_2 + \text{C}_3\text{H}_6 \rightarrow \text{C}_3\text{H}_5\text{X} + \text{HX}$
- $\text{C}_4\text{H}_8 \rightarrow \text{C}_4\text{H}_6 + \text{H}_2$
- more than one response is correct

_____ 70. Select the major product that would result from the reaction:



- $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- $\text{CH}_3\text{CH}_2\text{CH}_3$
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{SO}_4$

_____ 71. Select the major product that would result from the reaction:



- $\text{CH}_3\text{CH}(\text{Br})\text{CH}(\text{Br})\text{CH}_3$
- $\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
- $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{CH}_3$
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$

_____ 72. The conversion of cyclohexene to cyclohexane requires the reagent(s)

- HCl
- H_2O and H_2SO_4
- H_2 and H_2SO_4
- H_2 and Pt

_____ 73. The addition polymer produced from $\text{CH}_3-\text{CH}=\text{CH}-\text{Cl}$ is

- $\left[\text{CH}=\text{CH} \begin{array}{c} \text{CH}_3 \\ \text{Cl} \end{array} \right]_n$
- $\left[\text{CH}-\text{CH} \begin{array}{c} \text{CH}_3 \\ \text{Cl} \end{array} \right]_n$
- $\left[\text{CH}_2-\text{CH}=\text{CH}-\text{Cl} \right]_n$
- $\left[\text{CH}_2-\text{CH}=\text{CH}-\text{Cl} \right]_n$

_____ 74. The monomer of teflon (polytetrafluoroethene) is

- $\text{H}_2\text{C}=\text{CH}_2$
- $\text{H}_2\text{C}=\text{CF}_2$
- $\text{H}_2\text{C}=\text{CN}$
- $\text{CF}_2=\text{CF}_2$

____ 75. Which is the formula for an alkene?

- a. CH_3CHCH_2
- b. $\text{CH}_3\text{CH}_2\text{CH}_2$
- c. $\text{CH}_3\text{CH}_3\text{CH}_2$
- d. all of these.

____ 76. Which is the formula for an alkyne?

- a. $\text{CH}_3\text{CH}_2\text{CCH}_2$
- b. $\text{CH}_3\text{CH}_2\text{C}_2\text{CH}_3$
- c. $\text{CH}_3\text{CH}_2\text{CHCH}$
- d. $\text{CH}_3\text{CH}_2\text{CCH}_2$

____ 77. Which is a difference between butane and butene?

- a. butane burns and butene does not
- b. the presence of a double bond
- c. they are isomers
- d. the presence of a triple bond

____ 78. Which is a difference between butene and cyclobutene?

- a. They are isomers.
- b. Cyclobutene has 2 double bonds, butene does not.
- c. The phase variance of the infrared transconductance is asymmetric.
- d. Cyclobutene has fewer hydrogens than butene.

____ 79. Which is a difference between butyne and cyclobutyne?

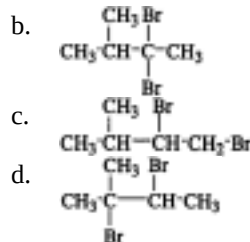
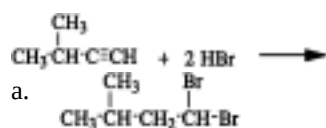
- a. Cyclobutyne can't exist!
- b. Butyne's multiple bond is not as strong as that of cyclobutyne.
- c. Cyclobutyne burns much hotter than butyne because of the greater unsaturation.
- d. Both b and c are differences between the molecules.

____ 80. The correct IUPAC name for the compound



- a. 4-bromopentyne
- b. 1-bromo-2-pentyne
- c. 1-bromo-3-pentyne
- d. 5-bromo-2-pentyne

____ 81. Select the product of the following reaction:



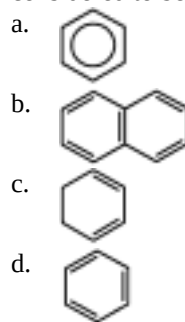
____ 82. Acetylene (ethyne gas) is commercially useful as:

- a. a starting material for plastics
- b. a fuel for torches
- c. an ingredient in pesticides
- d. a component in paint formulations

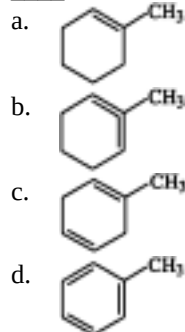
____ 83. What is the characteristic of aromatic compounds that is responsible for them being named aromatic compounds?

- a. The compounds have a pleasant smell.
- b. These compounds contain a benzene ring.
- c. It must contain a hydrocarbon chain that is either saturated or unsaturated and at least 3 carbons long.
- d. There is more than one correct response.

____ 84. Which of the following compounds is not considered to be aromatic?



____ 85. Which of the following structures is wrong?



Answer Section
TRUE/FALSE

1. T
2. T
3. F
4. T
5. F
6. F
7. T
8. F
9. F
10. T
11. F
12. T
13. T
14. F
15. T
16. F
17. T
18. T
19. F
20. T

21. T
22. T
23. T
24. F
25. T
26. T
27. T
28. T

MULTIPLE CHOICE

29. C
30. A
31. D
32. C
33. D
34. B
35. C
36. D
37. D
38. A
39. C
40. D
41. A

42. C
43. B
44. C
45. A
46. D
47. D
48. B
49. A
50. C
51. A
52. A
53. B
54. C
55. B
56. A
57. B
58. A
59. B
60. D
61. B
62. B
63. C

64. D
65. C
66. A
67. D
68. C
69. A
70. A
71. C
72. D
73. B
74. D
75. A
76. C
77. B
78. D
79. A
80. D
81. B
82. B
83. B
84. C
85. B